CONSULTING & BROKERAGES SERVICES

Project Alfie: A Theoretical Framework and Feasibility Analysis of a Staged, Hybrid Electromagnetic-Acoustic Direct-to-Orbit Launch System

Abstract

This paper presents a theoretical framework and critical feasibility analysis for Project Alfie, a novel ground-based launch system concept. The system is designed to propel a 3,000 kg payload to low Earth orbit using a hybrid propulsion method that synergistically combines staged electromagnetic acceleration with acoustic levitation and stabilization. The proposed architecture consists of a 50-meter vertical containment shaft housing a ten-stage modular accelerator. Each stage is designed to contribute a fraction of the total propulsive force required to achieve an orbital velocity of 7,800 m/s. Through a rigorous examination of the underlying physics and engineering requirements, this analysis quantifies the extreme operational parameters of the system. The findings reveal that the project's requirements for magnetic field strength, acoustic levitation scaling, payload survivability under acceleration, and peak power discharge exceed the current state-of-the-art by several orders of magnitude. These significant discrepancies define the boundaries of current and near-future technological capabilities and are presented as the primary challenges to the concept's viability. The paper concludes by identifying these feasibility gaps and proposing a roadmap for future research focused on fundamental physics and a necessary reconceptualization of the core design constraints.

I. Introduction: A Paradigm Shift in Non-Chemical Space Launch

1.1. The Imperative for New Launch Technologies

For over half a century, access to space has been almost exclusively reliant on chemical rocketry. While this technology has enabled monumental achievements, its foundational principles impose inherent limitations. Chemical rockets are characterized by low payload mass fractions, typically ranging from 2% to 4% of their total launch mass, resulting in high operational costs per kilogram delivered to orbit. Furthermore, the environmental impact of launching large volumes of propellants into the atmosphere is a growing concern. These factors provide a strong scientific and commercial impetus for the exploration of alternative, non-chemical space launch paradigms, particularly ground-based systems that could theoretically offer higher efficiency and lower recurring costs.

1.2. Conceptual Genesis of Project Alfie

Project Alfie began as a speculative inquiry into the nature of advanced propulsion, originating from the highly theoretical concept of creating a localized "time bubble" to isolate a user from their environment.[1] This initial exploration, rooted more in science fiction than in established physics, served as a catalyst for a structured process of scientific inquiry. The concept was systematically deconstructed and translated from its abstract origins into a tangible engineering problem grounded in fundamental physical principles. The investigation pivoted from speculative theories of spacetime manipulation involving general and special relativity to the concrete application of controllable forces.[1] This intellectual evolution led to the central challenge of the project: to design a system capable of generating and precisely controlling immense propulsive forces to counteract Earth's gravity and achieve orbital velocity. This process of refining a grand challenge into a series of solvable, albeit extremely demanding, technical problems forms the narrative of this paper.

1.3. The Hybrid Propulsion Hypothesis

This investigation culminates in the formulation of a hybrid propulsion hypothesis. It posits that by synergistically combining the immense acceleration potential of a staged electromagnetic coilgun with the stabilizing and supplemental forces of an acoustic levitation field, it is theoretically possible to propel a significant payload to orbital velocity within a highly constrained, ground-based infrastructure. The core of the proposed system is a vertical accelerator that uses a sequence of powerful, pulsed magnetic fields to drive the payload, while a surrounding field of high-intensity sound waves provides stabilization and counteracts instabilities. The primary objective of this paper is to rigorously test this hypothesis through detailed calculation, analysis of the required subsystems, and a critical feasibility assessment against the known limits of modern technology and physics.

II. The Electromagnetic Acceleration Module: A Critical Analysis

2.1. Foundational Principles

The primary propulsive force in the Project Alfie system is derived from electromagnetic principles, specifically the Lorentz force. This force is exerted on a current-carrying conductor within a magnetic field, providing a mechanism for non-contact acceleration. To generate the extraordinarily high forces required for orbital launch, the system design relies on high-temperature superconductors (HTS). Materials such as Yttrium Barium Copper Oxide (YBCO) are capable of carrying immense current densities with near-zero electrical resistance when cooled to cryogenic temperatures.[2] This property allows for the generation of magnetic fields far stronger than those achievable with conventional electromagnets, making HTS technology a cornerstone of the proposed accelerator design.[1]

2.2. Initial Calculations and the Feasibility Chasm

The fundamental challenge of the project is defined by the kinematic requirements of

launching a payload to orbit within a severely limited distance. The design specifies a 3,000 kg payload, a target orbital velocity (vf) of 7,800 m/s, and a launch shaft length (d) of only 50 meters.[1] Using the kinematic equation

vf2=v02+2ad, with an initial velocity (v0) of 0 m/s, the required acceleration (a) is calculated:

a=2dvf2=2×50 m(7800 m/s)2=100 m60,840,000 m2/s2=608,400 m/s2
This acceleration is immense. The total propulsive force (F) required to accelerate the 3,000 kg payload is given by Newton's second law, F=ma:

Ftotal=(3000 kg)×(608,400 m/s2)=1,825,200,000 N≈1.83×109 N
Initial calculations to generate this force using a single, continuous magnetic field resulted in requirements for magnetic field strength and energy delivery that were immediately identifiable as physically and technologically untenable.[1] This initial finding created a "feasibility chasm" that necessitated a fundamental redesign of the accelerator architecture.

2.3. Design Solution: The Ten-Stage Modular Accelerator

To address the impracticality of a single-stage system, the core design solution is a modular, ten-stage accelerator. The 50-meter containment shaft is divided into ten distinct 5-meter segments, each acting as an independent accelerator stage.[1] This architecture allows the total required propulsive force to be distributed across the modules. Each stage is responsible for generating a fraction of the total force, thereby reducing the instantaneous operational parameters for any single component.

- **Force per Stage:** Fstage=10Ftotal=101.825×109 N=1.825×108 N
- **Magnetic Field per Stage:** Based on the system's design parameters, the required magnetic field strength to produce this force within each stage is calculated to be **938 Tesla (T)**.[1]

2.4. Confronting the Magnetic Field Limit

The requirement of a 938 T magnetic field per stage represents the first major

technological barrier for Project Alfie. A review of the current state-of-the-art in magnet technology provides essential context. The world record for the most powerful *stable* magnetic field ever generated is approximately 45.22 T, achieved by a hybrid superconducting-resistive magnet at the Steady High Magnetic Field Facility (SHMFF) in China.[4] The National High Magnetic Field Laboratory (MagLab) in the United States has also achieved a stable field of 45 T and a brief, non-destructive peak of 45.5 T with a test magnet.[6]

This comparison reveals a profound gap: the Project Alfie requirement of 938 T is more than 20 times greater than the highest stable magnetic field ever produced on Earth. While pulsed magnets have generated fields exceeding 1,200 T, these events last for mere microseconds and are typically destructive to the magnet itself, rendering them unsuitable for a controlled propulsion application.[8] The project's feasibility is therefore contingent on a method to bridge this enormous performance gap.

2.5. Proposed Workaround: Magnetic Flux Concentrators (MFCs)

The proposed theoretical solution to this challenge is the implementation of Magnetic Flux Concentrators (MFCs). MFCs are structures made of high-permeability materials that guide and focus magnetic field lines, effectively amplifying the magnetic flux density in a specific region without increasing the power input to the source coil.[10] Research in sensor technology has shown that MFCs can enhance the sensitivity of magnetic sensors by factors of 10 to over 100, and in some specialized superconducting configurations, by factors exceeding 1,000.[12]

However, the application of MFCs in Project Alfie represents a significant leap from established practice. Current MFC research is overwhelmingly focused on micrometer- or millimeter-scale applications for enhancing sensor resolution.[13] There is no documented evidence of MFCs being successfully scaled to concentrate flux over the large volume of a launch tube to produce macro-scale propulsive forces with the required >20x gain. The physics of magnetic saturation in the concentrator material at such high field strengths and the engineering challenges of constructing such a device at this scale are major scientific unknowns.[16] Therefore, the reliance on MFCs to bridge the magnetic field gap must be framed not as a definitive solution, but as the project's single greatest point of scientific risk, requiring fundamental research

and experimental validation.

III. The Acoustic Levitation and Stabilization Module

3.1. Foundational Principles

In addition to the primary electromagnetic accelerator, the Project Alfie design incorporates an acoustic propulsion module. This system is based on the principle of acoustic radiation pressure, a phenomenon where high-intensity sound waves exert a physical force on an object.[17] By using an array of ultrasonic transducers and a reflector, it is possible to create a standing wave—a stationary pattern of high and low pressure zones. Objects can be trapped and suspended in the low-pressure nodes of this standing wave, a process known as acoustic levitation.[1] The design hypothesizes that this effect can be used to provide both supplemental lift and, more critically, active stabilization for the payload as it accelerates through the launch tube.

3.2. Calculation of Acoustic Pressure Requirements

The system is designed to operate in a staged manner, with the acoustic module contributing a varying percentage of the required lifting force in coordination with the electromagnetic accelerator. The calculations indicate a maximum required acoustic pressure of **10,292.14 Pascals (Pa)** during the third stage of the launch sequence.[1] This pressure must be generated and sustained over the surface area of the 3,000 kg payload to exert the intended force.

3.3. Confronting the Mass-Scaling Limit

The proposal to use acoustic forces to lift a 3,000 kg (3,000,000 g) payload

confronts a feasibility gap even more profound than that of the magnetic system. The current documented world record for the heaviest object ever acoustically levitated is a 0.6-gram expanded polystyrene sphere.[17] This represents a mass gap of a factor of 5 million between the project's requirement and demonstrated physical reality.

This disparity is not merely an engineering challenge but a confrontation with fundamental scaling laws. The force exerted by an acoustic field is proportional to the surface area over which it acts (a squared term), while the gravitational force it must overcome is proportional to the object's mass, and thus its volume (a cubed term). This "square-cube law" makes it exceptionally difficult to scale acoustic levitation to large, dense objects.[19] The immense power required to generate the necessary acoustic pressure over a large area, and the inherent instabilities that arise when trying to trap objects larger than the sound's wavelength, are significant physical barriers.[18]

Given this vast disparity, the role of the acoustic system as a primary lifting mechanism is highly speculative. A more physically plausible application would be to re-contextualize its function. The small, precise, and rapidly controllable forces generated by phased acoustic arrays are well-suited for fine-grained stabilization and attitude control. In this capacity, the acoustic module could serve to counteract minor oscillations and keep the payload perfectly centered within the magnetic field, a critical function for a high-velocity launch system. This reframing makes the acoustic component more credible, shifting its purpose from brute-force lifting to high-precision guidance.

3.4. Proposed Workaround: Resonant Chamber Amplification

To address the challenge of generating the required high-intensity sound, the design incorporates the use of resonant chambers.[1] A resonant chamber is a cavity that uses constructive interference to amplify sound waves at its natural resonant frequencies.[21] By confining and reflecting the sound waves, the chamber allows energy from the transducers to accumulate, increasing the intensity of the standing wave without a proportional increase in input power. In some acoustic systems, resonance can amplify pressure waves by factors of up to 50.[23] The design proposes using an array of transducers coupled with resonant chambers to achieve the target pressure of 10,292.14 Pa more efficiently.[1] While this approach is sound in principle, it does not

fundamentally overcome the mass-scaling limitations previously discussed.

IV. Integrated System Dynamics and Vehicle Architecture

4.1. The Staged Force Application Strategy

The Project Alfie launch sequence is designed as a synchronized operation between the electromagnetic and acoustic modules, applying force incrementally across four distinct phases. This strategy aims to manage the immense energy transfer and maintain payload stability throughout the brief launch window.[1]

- **Stage 1: Initial Lift and Stability.** Both systems contribute 10% of the total required force, generating a combined force of approximately 3.65×108 N to initiate upward movement and establish a stable levitation field.
- **Stage 2: Increased Lift.** The contribution from both systems increases to 15% each, raising the cumulative force to overcome the initial inertia and begin significant acceleration.
- **Stage 3: Intermediate Lift.** The systems ramp up to their maximum designed output, each contributing 20% of the total force, delivering the peak acceleration of the launch sequence.
- **Stage 4: Final Lift and Acceleration.** The force contribution is reduced back to 10% from each system for the final push to orbital velocity, likely for fine-tuning the exit trajectory.

4.2. Vehicle Design and Structural Analysis

The payload is housed within a dedicated vehicle designed for the unique rigors of this launch method. The proposed architecture is a 3D cone-shaped vehicle, a geometry chosen for its inherent aerodynamic stability during high-velocity vertical ascent.[1] The structure would be fabricated from advanced materials with high specific strength, such as carbon fiber reinforced polymers (CFRP) and high-strength steel or

titanium alloys, to withstand the extreme forces involved.[1]

4.3. Confronting the G-Force Limit

The most severe constraint imposed by the system's design is the extraordinary level of acceleration. As calculated, the vehicle and its payload must endure an acceleration of **608,400 m/s²**. To put this into perspective, this value can be converted to g-force, where 1 g is the standard acceleration due to gravity at Earth's surface (9.81 m/s2).

g-force=9.81 m/s2608,400 m/s2≈62,018 g
This level of acceleration is physically destructive and represents a fundamental flaw in the design if the goal is to launch conventional payloads.

- **Material Tolerance:** While advanced materials like CFRP and maraging steels possess very high strength-to-weight ratios, subjecting any complex mechanical structure to over 60,000 g would impose stresses likely to cause catastrophic structural failure.[24] The forces involved would exceed the material's ability to transmit them without deformation or fracture.
- **Electronics Survivability:** Hardening electronics for high-g environments is a specialized field, particularly for military applications like artillery shells. While some components can be designed to survive shocks up to 50,000 g, this is at the absolute limit of current technology and typically applies to simple, robust, and often potted electronic assemblies.[28] A complex guidance or communication system would not survive.
- **Payload Survivability:** This is the most critical point. Any conventional cargo—be it satellites, supplies, or biological material—would be instantly destroyed. A 1 kg object would experience a force equivalent to the weight of over 62,000 kg. This renders the system as currently designed viable only for launching highly robust, solid blocks of inert, durable material.

The immutable physics of the kinematic equation a=vf2/(2d) reveals that this destructive g-force is a direct and unavoidable consequence of the 50-meter launch shaft constraint. To achieve orbital velocity in such a short distance, this extreme acceleration is a mathematical necessity. This constraint is therefore a fundamental design flaw that makes the system impractical for almost any conceivable space launch application.

Table 1: Final Vehicle and Launch Parameters

Parameter	Unit	Value
Payload Mass	kg	3,000
Vehicle Shape	-	3D Cone
Target Velocity	m/s	7,800
Launch Shaft Length	m	50
Total Required Acceleration	m/s^2	608,400
Equivalent G-Force	g	~62,018
Total Propulsive Force	N	1.825×109

V. Critical Sub-System Analysis: Power and Thermal Management

5.1. The Proposed Power Architecture

To meet the energy demands of the system, the design proposes a hybrid power architecture. This system would combine high-energy-density lithium-ion batteries for sustained storage with high-power-density supercapacitors.[1] Supercapacitors are capable of extremely rapid charge and discharge cycles, making them theoretically suitable for delivering the intense bursts of power required to energize the electromagnetic coils in sequence.[30] The Li-ion batteries would serve to recharge the supercapacitor banks between launches.[32]

5.2. Re-evaluating the Power Requirement

The analysis documented in the project's development contains a critical miscalculation of the power requirement that fundamentally alters the assessment of its feasibility. The project's calculations estimate a total system power of **131 kW**.[1] This figure appears to be based on the electrical power dissipated by the coils as resistive losses (

P=IV). However, this neglects the primary purpose of the system: performing work on the payload.

The true power requirement must be calculated based on the rate at which kinetic energy is imparted to the 3,000 kg vehicle.

- Total Kinetic Energy (KE): The final kinetic energy of the payload upon exiting the shaft is:

  KE=21mvf2=21(3000 kg)(7800 m/s)2=9.126×1010 Joules
- Launch Time (t): The total time the vehicle spends accelerating in the 50-meter shaft is:

  t=a2d=608,400 m/s22×50 m=0.00016436 s2≈0.0128 seconds
- Average Power (Pavg): The average power delivered to the payload during this time is:

  Pavg=tKE=0.0128 s9.126×1010 J≈7.13×1012 Watts=7.13 Terawatts (TW)
- Peak Power (Ppeak): The instantaneous power peaks at the moment of exit, when the vehicle is at maximum velocity. Power is the product of force and velocity (P=Fv):

  Ppeak=Ftotal×vf=(1.825×109 N)×(7800 m/s)≈1.42×1013 Watts=14.2 Terawatts (TW)

This corrected calculation reveals a staggering discrepancy. The actual peak power requirement of 14.2 TW is more than 100 million times greater than the 131 kW figure calculated in the project's development.

5.3. Confronting the Power Limit

A peak power requirement on the terawatt scale places Project Alfie far beyond the capabilities of any existing or foreseeable energy storage technology. The total installed electricity generation capacity of the entire world is in the range of a few terawatts. The project would require a peak power draw equivalent to that of multiple continents combined, delivered in a fraction of a second. The proposed battery and supercapacitor system is orders of magnitude insufficient for this task.[30] This miscalculation represents the most significant and insurmountable feasibility flaw in the entire concept. It reframes the primary challenge from one of device engineering to one of national- or global-scale energy infrastructure.

Table 2: Staged Propulsion Profile and Corrected Power Requirements

Stage #	Force (MN)	Magnetic Field (T)	Acoustic Pressure (kPa)	Peak Power (TW)
1	182.5	938	7.28	3.56
2	273.8	938	8.91	6.50
3	365.0	938	10.29	9.87
4	182.5	938	7.28	14.24
...	...	...	...	...
10	182.5	938	7.28	14.24
Peak	**365.0**	**938**	**10.29**	**14.24**

(Note: Peak power is calculated based on the velocity at the end of each conceptual force application phase, culminating in the final peak power at exit velocity.)

5.4. Thermal Management Systems

The design correctly identifies the need for robust thermal management, including liquid nitrogen cooling for the HTS coils and active cooling for transducers and power electronics.[1] Given the corrected terawatt-scale power requirements, the challenge of

dissipating waste heat becomes monumental. Even with the near-perfect efficiency of superconductors, the sheer magnitude of energy flowing through the system would generate enormous thermal loads on all components, requiring an industrial-scale cryogenic and cooling infrastructure.

VI. Comprehensive Feasibility Assessment and Future Outlook

6.1. Synthesis of Key Feasibility Gaps

The preceding analysis has identified four critical and quantifiable gaps between the requirements of the Project Alfie concept and the current state of science and technology. These gaps are not incremental but represent differences of multiple orders of magnitude.

1. **The Magnetic Field Gap:** The requirement for a 938 T stable magnetic field per stage is over 20 times greater than the world record of ~45 T. The proposed solution, Magnetic Flux Concentrators, is unproven for this scale and application.
2. **The Acoustic Scaling Gap:** The requirement to levitate a 3,000,000 g payload is 5 million times greater than the 0.6 g record, challenging the fundamental scaling laws of acoustic physics.
3. **The G-Force Survivability Gap:** The ~62,000 g acceleration, a direct result of the 50-meter launch constraint, would destroy any conventional payload and likely the launch vehicle itself. This exceeds the survivability limits of even hardened military electronics.
4. **The Power Requirement Gap:** The corrected peak power requirement of 14.2 TW is over 100 million times greater than the initial project estimate and rivals the combined power generation capacity of entire continents.

Table 3: Feasibility Gap Analysis

Performance Metric	Project Alfie Requirement	Current State-of-the-Art	Performance Gap (Factor)

Stable Magnetic Field	938 T	~45 T [4]	~20.8x
Max. Acoustically Levitated Mass	3,000,000 g	0.6 g [17]	5,000,000x
G-Force (Payload Survivability)	~62,018 g	~50,000 g (Hardened Electronics) [29]	>1.2x (for electronics); >>100x (for structures)
Peak Power Discharge	14.2 TW	~GW-scale (Large Pulsed Power Facilities)	>10,000x

6.2. Conclusive Assessment of Viability

Based on this comprehensive, evidence-based analysis, the Project Alfie concept as currently designed is not feasible. Its operational parameters exceed the known limits of materials science, magnet technology, acoustic physics, and energy systems not by incremental margins that might be overcome with optimization, but by orders ofmagnitude. Each of the four identified feasibility gaps represents a barrier that would require multiple fundamental breakthroughs in science and engineering to surmount.

6.3. Recommendations for Future Research

While the current design is untenable, the intellectual exercise of Project Alfie provides a valuable framework for identifying the frontiers of advanced propulsion research. The identified challenges can be transformed into a constructive roadmap for future inquiry.

- **Recommendation 1: Redesign the Fundamental Launch Profile.** The most critical and impactful change would be to abandon the 50-meter launch shaft constraint. The destructive g-forces and the terawatt-level power requirements are both direct consequences of this single parameter. By extending the acceleration path to several kilometers (akin to existing maglev tracks or

proposed space guns), the required acceleration and peak power would decrease dramatically, potentially bringing them into a realm of engineering possibility. This single change would alleviate the g-force and power gaps simultaneously.

- **Recommendation 2: Initiate Fundamental Research into Scalable Force Fields.** The project highlights the need for fundamental, experimental research in two key areas. First, investigating the potential for scaling Magnetic Flux Concentrators for macro-scale propulsive applications, moving beyond their current use in sensors. Second, exploring the physics of high-mass, high-power acoustic fields to determine if there are any viable pathways, perhaps involving novel metamaterials, to overcome the square-cube law limitation for stabilization or partial lift applications.
- **Recommendation 3: Explore Next-Generation Pulsed Power Systems.** The power challenge necessitates looking beyond conventional battery and supercapacitor technology. Future research should focus on developing pulsed power systems capable of terawatt-level discharge. This domain includes technologies such as large-scale capacitor banks, magnetohydrodynamic generators, and other high-energy-density physics concepts that could, in the long term, provide the instantaneous power required for such an ambitious undertaking.

Works cited

1. alfie raw data v1.txt
2. Intro to high-temperature superconductors - Magnet Academy - National MagLab, accessed on July 23, 2025, https://nationalmaglab.org/magnet-academy/read-science-stories/science-simplified/high-temperature-superconductors/
3. BEST Reports Improved In-Field Critical Current Performance of its Second Generation High-Temperature Superconductor YBCO Tapes - Bruker Investor Relations, accessed on July 23, 2025, https://ir.bruker.com/press-releases/press-release-details/2010/BEST-Reports-Improved-In-Field-Critical-Current-Performance-of-its-Second-Generation-High-Temperature-Superconductor-YBCO-Tapes/default.aspx
4. Scientists Just Broke The World Record For The Most Powerful Stable Magnetic Field, accessed on July 23, 2025, https://www.sciencealert.com/scientists-just-broke-the-world-record-for-the-most-powerful-stable-magnetic-field
5. World's strongest steady magnetic field generated in China - New Atlas, accessed on July 23, 2025, https://newatlas.com/physics/worlds-strongest-steady-magnetic-field/
6. With mini magnet, National MagLab creates world-record magnetic field, accessed on July 23, 2025,

https://nationalmaglab.org/news-events/news/lbc-project-world-record-magnetic-field/
7. en.wikipedia.org, accessed on July 23, 2025, https://en.wikipedia.org/wiki/Bitter_electromagnet
8. What Is the Strongest Magnet, accessed on July 23, 2025, https://www.stanfordmagnets.com/what-is-the-strongest-magnet.html
9. Japanese Physicists Generate Strongest Magnetic Field Ever Achieved Indoors | Sci.News, accessed on July 23, 2025, https://www.sci.news/physics/strongest-magnetic-field-achieved-indoors-06420.html
10. Discretely structured magnetic flux concentrators - arXiv, accessed on July 23, 2025, https://arxiv.org/html/2506.01424v1
11. Magnetic Flux Concentrators for Induction Heating - Field Concentrator Coil Production | Induction Technology Solutions, Inc., accessed on July 23, 2025, https://inductronix.com/magnetic-flux-concentrators/
12. Magnetic Flux Concentration Technology Based on Soft Magnets and Superconductors, accessed on July 23, 2025, https://www.mdpi.com/2073-4352/14/8/747
13. Magnetic flux concentration at micrometer scale - HKU EEE, accessed on July 23, 2025, https://www.eee.hku.hk/~ppong/manuscripts/magnetic%20flux%20concentration%20at%20micrometer.pdf
14. Double-Gap Magnetic Flux Concentrator Design for High-Sensitivity Magnetic Tunnel Junction Sensors - MDPI, accessed on July 23, 2025, https://www.mdpi.com/1424-8220/19/20/4475
15. Design of magnetic flux concentrator composed of nanospheres for high-sensitivity magnetometers - AIP Publishing, accessed on July 23, 2025, https://pubs.aip.org/aip/jap/article/136/17/173102/3318531/Design-of-magnetic-flux-concentrator-composed-of
16. MATERIAL SELECTION AND GEOMETRY CONSIDERATIONS FOR HIGH CURRENT SENSING MAGNETIC FLUX CONCENTRATOR DESIGNS - Allegro MicroSystems, accessed on July 23, 2025, https://www.allegromicro.com/-/media/files/application-notes/an296245-high-current-sensing-flux-concentrator-design.pdf?sc_lang=en
17. Acoustic levitation - Wikipedia, accessed on July 23, 2025, https://en.wikipedia.org/wiki/Acoustic_levitation
18. How Acoustic Levitation Works - Science | HowStuffWorks, accessed on July 23, 2025, https://science.howstuffworks.com/acoustic-levitation.htm
19. Dumb question: Can we use sound waves for large scale levitation? - Reddit, accessed on July 23, 2025, https://www.reddit.com/r/askscience/comments/3pq3l6/dumb_question_can_we_use_sound_waves_for_large/
20. Scientists Just Used a Tractor Beam to Levitate the Largest Object Yet - YouTube, accessed on July 23, 2025, https://www.youtube.com/watch?v=gHAe4FFHtB0
21. Resonance chamber - Wikipedia, accessed on July 23, 2025,

https://en.wikipedia.org/wiki/Resonance_chamber
22. Acoustic resonance - Wikipedia, accessed on July 23, 2025, https://en.wikipedia.org/wiki/Acoustic_resonance
23. Acoustical resonance - YouTube, accessed on July 23, 2025, https://www.youtube.com/watch?v=AmRydYHfXnk
24. Carbon fibers - Wikipedia, accessed on July 23, 2025, https://en.wikipedia.org/wiki/Carbon_fibers
25. Evaluation of Steel Alloys Subjected to High-Frequency Flexural Motion. - DTIC, accessed on July 23, 2025, https://apps.dtic.mil/sti/tr/pdf/ADA051363.pdf
26. Specific strength - Wikipedia, accessed on July 23, 2025, https://en.wikipedia.org/wiki/Specific_strength
27. g-force - Wikipedia, accessed on July 23, 2025, https://en.wikipedia.org/wiki/G-force
28. High-G Survivability - Auburn University, accessed on July 23, 2025, https://eng.auburn.edu/auepri/facilities/highG.html
29. SURVIVABILITY ASSESSMENT AND LIFE-MODELING OF FINE PITCH SOLDER JOINT FUZE ELECTRONICS UNDER MECHANICAL SHOCK LOADS UP TO 50,000 - Circuit Insight, accessed on July 23, 2025, https://www.circuitinsight.com/pdf/Survivability_Assessment_Life-Modeling_Fine_Pitch_Solder_Joint_Fuze_Electronics_Under_Mechanical_Shock_Loads_Up_%2050000G_smta.pdf
30. Advanced supercapacitor sets energy density record at 418 Wh/kg - Rinnovabili, accessed on July 23, 2025, https://www.rinnovabili.net/tech-innovations-and-startups/energy-tech-innovations-and-startups/advanced-supercapacitor-energy-density-record/
31. Ultracapacitor technology | Skeleton, accessed on July 23, 2025, https://www.skeletontech.com/ultracapacitor-technology
32. Lithium ion batteries: energy density?, accessed on July 23, 2025, https://thundersaidenergy.com/downloads/lithium-ion-batteries-energy-density/
33. Lithium-Ion Battery - Clean Energy Institute - University of Washington, accessed on July 23, 2025, https://www.cei.washington.edu/research/energy-storage/lithium-ion-battery/
34. Comparative analysis of lithium-ion and flow batteries for advanced energy storage technologies - MATEC Web of Conferences, accessed on July 23, 2025, https://www.matec-conferences.org/articles/matecconf/pdf/2024/04/matecconf_icmed2024_01176.pdf